3. Classify the order of convergence of the following sequences:

(a) $a_n = 2^{-n}$.

(b) $a_n = (n+1)^{-1}$.

(c) $a_n = 2^{-2^n}$.

For each sequence, plot a_n from $n = 1$ up to N, where $N = 100$ for (a) and (b) and $N = 10$ for (c), using a logarithmic scale on the y-axis.

a) $\lim_{n\to\infty} \frac{2^{-(n+1)}}{2^{-n}} = \frac{1}{2} \in (0,1)$ so 2^{-n} is linear.

b) $\lim_{n\to\infty} \frac{((n+1)+1)^{-1}}{((n+1)-1)} = \lim_{n\to\infty} \frac{1+\frac{1}{n}}{1+\frac{2}{n}} = 1$ so $\frac{1}{n+1}$ is sublinear.

c) $\lim_{n\to\infty} \frac{2^{-2^{(n+1)}}}{2^{-2^n}} = \lim_{n\to\infty} \frac{2^{2^n}}{(2^{2^n})^2} = 0$. $\lim_{n\to\infty} \frac{2^{2^{-(n+1)}}}{(2^{2^{-n}})^q} = 1$ if $q=2$ so $2^{2^{-n}}$ is quadratic.

4. Does Banach's contraction mapping theorem apply to the following continuous real-valued functions g with domain of definition X? Justify your answers.

(a) $X = \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$, $\quad g(x) = \sin x + x$.

(b) $X = [0, 2]$, $\quad g(x) = x^3 + 3x^2 - 3$.

(c) $X = [1, \infty)$, $\quad g(x) = x + x^{-1}$.

a) $|g'(x)| = |\cos x + 1| < 1 \quad \forall x \in \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$ as $\cos x \in \left[-1, \frac{-\sqrt{2}}{2}\right]$ for $x \in \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$.
$|g'(x)| \leq 1 - \frac{\sqrt{2}}{2} < 0.9 < 1$, $0.9 \in [0,1)$ so Banach's contraction mapping theorem applies.

b) $g(0) = -3$ so we do not have $g: X \to X$. Hence, Banach's contraction mapping theorem doesn't apply.

c) $|g(x) - g(y)| = \left|x - y + \frac{1}{x} - \frac{1}{y}\right| \leq |x-y| + \left|\frac{1}{x} - \frac{1}{y}\right|$ by triangle inequality. There does not exist $0 \leq \gamma < 1$ s.t. $|x-y| + \left|\frac{1}{x} - \frac{1}{y}\right| \leq \gamma |x-y|$ as $\left|\frac{1}{x} - \frac{1}{y}\right| > 0$. Thus Banach's contraction mapping theorem doesn't apply.

5. (a) Let $a_n \to a$ be a convergent sequence of real numbers. Let $e_n := |a_n - a|$ be its associated sequence of errors. Assume that the error can be described as

$$e_{n+1} \approx M e_n^q$$

for constants M and q.

Given three successive terms e_n, e_{n+1}, e_{n+2} in the sequence of errors, describe a procedure to estimate M and q.

(b) Consider the Leibniz formula for π:

$$\pi = 4 \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1},$$

and let a_n be its sequence of partial sums.

Estimate M and q for this sequence, starting from e_n with $n = 10,000$, and thereby state a conjecture about its convergence order.

(c) Recall that the Aitken acceleration $\tilde{a}_n$ of a sequence a_n is given by

$$\tilde{a}_n = a_n - \frac{(a_{n+1} - a_n)^2}{a_{n+2} - 2a_{n+1} + a_n}.$$

Plot the sequence of errors up to $n = 10,000$ of the partial sums a_n of the Leibniz formula for π, as well as its Aitken acceleration $\tilde{a}_n$, and the iterated Aitken acceleration $\tilde{\tilde{a}}_n$.

(d) What is the estimated order of convergence of $\tilde{a}_n$ and $\tilde{\tilde{a}}_n$? Comment on your observations.

[*Hint: you may wish to use higher-precision arithmetic for this question. Use* `mpmath` *with* `mpmath.mp.dps = 100`.]

a) $\log e_{n+1} \approx \log M + q \log e_n$, $\log e_{n+2} \approx \log M + q \log e_{n+1}$. Therefore, calculate $q \approx \frac{\log e_{n+2} - \log e_{n+1}}{\log e_{n+1} - \log e_n} = \frac{\log\left(\frac{e_{n+2}}{e_{n+1}}\right)}{\log\left(\frac{e_{n+1}}{e_n}\right)}$ and $M \approx \frac{e_{n+2}}{e_{n+1}^q}$.

b) $q \to 1$ and $M \to 1$ so $e_{n+1} \approx M e_n \Rightarrow e_n$ is linear.

d) $\tilde{a}_n$ has quadratic terms of a linear sequence so has quadratic order. $\tilde{\tilde{a}}_n$ has cubic order as change of order from a_n to $\tilde{a}_n$ must be by the same factor as $\tilde{a}_n$ to $\tilde{\tilde{a}}_n$.